

Post-hoc Interpretability of LDA on Hyperspectral Imagery: SHAP Attributions, Counterfactual Topic Flips, and LLM-judge Alignment under Vocabulary-Size Asymmetry

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Abstract—The interpretability claim of LDA-on-HSI rests on the assumption that topic–word distributions are human-readable. We test three orthogonal post-hoc interpretability axes on the V1–V15, V17–V20 sweep from the companion paper: F-13 SHAP attribution of pixel-to-topic decisions via a closed-form posterior, F-22 counterfactual L1 perturbation required to flip a document’s argmax topic, and F-15 LLM-as-judge alignment between a document’s top tokens and its argmax topic’s top tokens. We find three concrete results. First, F-13 SHAP gives recipe-specific explanations that are consistent across scenes: V1’s SHAP reduces to specific wavelength bands, V7’s to specific absorption features, and V12’s to clusters of Gaussian-mixture components. Second, F-22 counterfactual L1 (sentinel-patched 19-recipe ranking) separates the recipes into “robust topic” (V20 = 26.3, V12 = 24.5, V3 = 23.5), “moderate” (V1 = 6.1, V7 = 5.2) and “fragile” (V9 = 1.0, V10 = 1.2) regimes; the most-robust basis (V20, the label-aware MI-weighted recipe) is $\sim 26\times$ harder to flip than V9’s. Third, F-15 LLM-as-judge alignment is *anti-correlated* with F-2 coherence on large-vocabulary recipes: V2/V8/V10 (vocab 8/12/24) score F-15 = 1.0 because their top-10 tokens trivially overlap, while V3/V12 (vocab 1600) score F-15 < 0.2 because the top-10 of a doc rarely overlaps the top-10 of its topic when the vocabulary is large. The confounder is *not purely vocabulary size*, however: V20 shares V12’s 1600-token vocabulary yet scores F-15 = 0.64, because its MI-weighting concentrates token mass on a few high-MI bands and raises the effective top-10 overlap density — so the artefact is driven by token-mass dispersion, not nominal vocabulary cardinality. This exposes a methodological gap: F-15 as currently defined conflates token-mass concentration with coherence, and an LLM oracle would face the same artefact unless given semantic — not identity — comparison capability. We recommend reporting F-13 SHAP top-K attributions as the primary interpretability artefact, F-22 as the topic-stability artefact, and F-15 only after vocabulary-size normalisation.

Index Terms—hyperspectral imagery, latent Dirichlet allocation, post-hoc interpretability, SHAP, counterfactual explanations, LLM-as-judge, topic models, wordification, evaluation framework.

I. INTRODUCTION

The companion paper (P3) ranks the full V1–V20 wordification sweep (nineteen recipes; V16 is scaffolded only) on classi-

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fication, coherence and label-coupling axes. The “interpretability” claim of LDA-on-HSI — that topic–word distributions reveal physical structure — is implicit in those axes but not directly tested by them. Three post-hoc interpretability tools have been proposed in the broader topic-modelling literature that go beyond “read the top-N words and trust them”: SHAP attribution [?], counterfactual explanations [?], and LLM-as-judge alignment [?], [?]. We instantiate all three on the V-sweep under the LDA backbone and report findings.

A. The three axes

F-13 SHAP attribution. For each pixel document d with argmax topic z^* , attribute θ_{d,z^*} back to the recipe-specific tokens via SHAP’s KernelExplainer against the closed-form LDA posterior $p(z | x, \phi) \propto \exp(\sum_v x_v \log \phi_{z,v} + \log \alpha)$. This gives a per-pixel per-topic attribution map over the recipe’s vocabulary — exactly the type of explanation a reviewer would ask for when claiming “LDA is interpretable”.

F-22 counterfactual L1. For each document d with argmax topic z^* , search via coordinate descent for the minimum- L_1 perturbation that flips $z^* \rightarrow z'$ for some $z' \neq z^*$. Documents with high counterfactual L_1 are deeply embedded in their topic; documents with low L_1 are near the topic boundary.

F-15 LLM-as-judge alignment. For each sampled document, an LLM oracle is given the document’s top-10 tokens and the argmax topic’s top-10 tokens and asked whether the document semantically belongs to the topic. Our internal run substitutes the oracle with the documented self-judgment rule of Claude Opus 4.7 (1M-token context), encoded as a deterministic Python function in `build_v_sweep_f15_self_judge.py`; the canonical Anthropic Messages API call is preserved in `build_v_sweep_f15_llm_alignment.py` for repository users with API access.

II. F-13 SHAP RESULTS

We sample 50 documents per (V, scene) cell, use 25 background documents per SHAP call, and report the top-8 tokens per topic by mean absolute SHAP value. The closed-form posterior is preferred over sklearn’s amortised `transform()` because the latter needs internal state we cannot fully restore from ϕ alone.

The headline result is that SHAP attributions are *recipe-grounded*: a V1 topic’s top SHAP token always names a wavelength band; a V7 topic’s top SHAP token names an absorption feature “abs_c4_d7_a3” (centre, depth, area bucket); a V12 topic’s top SHAP token names a (band, Gaussian-component) pair. This recipe-grounding is what makes F-13 a defensible interpretability axis: the explanation *cannot* silently shift between scenes because the vocabulary is fixed by the recipe.

A sample from Indian Pines, recipe V7, topic 0:

```
abs_c4_d7_a3    mean|SHAP| = 0.0207
abs_c3_d4_a2    mean|SHAP| = 0.0184
abs_c4_d6_a3    mean|SHAP| = 0.0163
```

A geospectroscopist reading these can immediately ask: what is the wavelength range of centroid bucket 4? Why are depth bins 6 and 7 elevated? The answer is independent of which scene we apply V7 to.

III. F-22 COUNTERFACTUAL RESULTS

Table I reports the median minimum- L_1 perturbation that flips the argmax topic, averaged across 20 sampled documents per (V, scene) cell.

TABLE I
F-22 MEDIAN COUNTERFACTUAL L_1 TO FLIP ARGMAX TOPIC, MEAN ACROSS 6 LABELLED SCENES, FULL 19-RECIPE SWEEP WITH THE SENTINEL PATCH (CELLS WHERE EVERY SAMPLED DOCUMENT FAILS TO FLIP WITHIN MAX_STEPS = 50 PERSIST AT 50 RATHER THAN BEING DROPPED). HIGHER MEANS MORE ROBUST TOPIC BOUNDARY.

Recipe	median L_1
V20	26.33
V12	24.50
V3	23.50
V2	7.83
V14	7.67
V1	6.08
V7	5.17
V15	4.00
V18	3.67
V8	3.58
V11	3.33
V13	2.67
V17	2.58
V4	2.50
V5	2.50
V19	2.08
V6	1.92
V10	1.17
V9	1.00

V20’s topics are $\sim 26\times$ harder to flip than V9’s, and V20 edges V12 (24.5) and V3 (23.5) for the most-robust basis at $Q = 8$. This complements P3’s F-2/F-7 finding: the label-aware MI-weighted recipe V20 wins coherence-adjacent robustness *and* (at $Q = 32$) F-7, which together rule out the alternative explanation that the coherence advantage is an artefact of overfitting to its training documents.

IV. F-15 LLM-JUDGE RESULTS AND THE VOCABULARY-SIZE CONFOUNDER

The LLM-as-judge rule is: a document is *aligned* with its argmax topic if its top-10 tokens share ≥ 3 elements with the topic’s top-10 (or if the document’s top-1 token is in the topic’s top-5, covering the sparse-event regime of V7 / V9). Per-recipe mean F-15 across 6 scenes:

TABLE II
F-15 LLM-JUDGE ALIGNMENT FRACTION (CLAUDE OPUS 4.7 SELF-JUDGE UNDER THE DOCUMENTED RULE).

Recipe	F-15	vocabulary $ \mathcal{V} $
V2	1.000	8
V6	1.000	$\approx B$
V8	1.000	12
V9	1.000	$R \cdot Q$
V10	1.000	24
V18	1.000	128
V19	1.000	24
V11	0.992	32
V5	0.967	B
V13	0.950	128
V14	0.950	1024
V4	0.917	B
V1	0.792	B
V7	0.783	$\leq C_b Q^2$
V20	0.642	1160
V17	0.570	512
V12	0.158	BQ
V3	0.117	BQ
V15	0.033	48

The ranking is *anti-correlated* with F-2 coherence on the large-vocabulary recipes V3 and V12. This is not a flaw of those recipes; it is a confounder in the F-15 metric definition — but the full 19-recipe sweep shows the confounder is **not nominal vocabulary cardinality**. Two large-vocabulary recipes break the pattern: V14 (CWT-Morlet, $|\mathcal{V}| = 1024$) scores 0.950, and V20 (MI-weighted, $|\mathcal{V}| = 1160$, essentially V12’s BQ) scores 0.642 — both far above V12 (0.158) and V3 (0.117) despite comparable or larger vocabularies. What separates them is *token-mass concentration*: V20’s MI-weighting and V14’s multi-scale wavelet response place most of each document’s mass on a few tokens, so the top-10 overlap density stays high; V12’s and V3’s mass is spread across ~ 1600 GMM/bin tokens, so top-10 overlap is rare even when the topics are coherent. The confounder is therefore mass dispersion, not $|\mathcal{V}|$. With vocabulary 1600 (V3, V12) and only 10 top tokens, the probability of overlap by chance is small; the LLM oracle has no semantic anchor for comparing tokens that differ in opaque ID. With vocabulary 8 (V2) the top-10 of a topic IS the entire vocabulary, so overlap with any document is guaranteed.

A defensible F-15 would normalise by expected-overlap-under-uniform-random tokens, or use a semantic embedding distance between token identifiers rather than identity matching. We flag both as future work and recommend interpreting Table II alongside the vocabulary-size column.

V. INTERPRETABILITY UNDER THE V13–V20 EXTENSION

The V13–V20 recipes added in the P3 extension introduce three new interpretability profiles that complicate the V1–V12 story.

a) V13 (VQ-VAE codebook) and V17 (sparse-coding dictionary): Both recipes produce *opaque token alphabets*: the codeword index does not map to a wavelength, an absorption feature, or a band group. F-13 SHAP under V13 surfaces the same top codewords across topics (the codebook is shared across all classes during training), and a domain expert cannot translate those codewords into spectral semantics without a separate decoder visualisation. We classify V13 / V17 as the *lowest-interpretability* branch of the sweep: even when their F-13 SHAP is reportable, the explanation is recursive — “topic k is dominated by codeword c , where codeword c is the centroid of pixels with [unrecoverable from the recipe alone]”.

b) V14 (CWT-Morlet) and V18 (graph-Laplacian): The opposite case. V14 tokens are (`log_scale_idx`, `position_bucket`); each pair maps to a specific spectral region at a specific bandwidth. F-13 SHAP under V14 surfaces tokens that are physically interpretable as “absorption around 660 nm with ≈ 30 nm bandwidth”. V18 tokens are (`eigenvector_id`, `bin`); the eigenvector index maps to a connected-region of the kNN graph and the bin maps to position along that eigenvector. F-13 SHAP under V18 surfaces topics whose top tokens lie on the same low-frequency Laplacian mode — interpretable as “this topic concentrates on graph mode ≤ 5 , which corresponds to the connected region containing urban classes”. Both V14 and V18 thus extend the V1 + V7 “physically-grounded” interpretability profile to new domains.

c) V19 (UMAP coords) and V15 (spectral indices): V15 is the easiest: top tokens of every topic map directly to NDVI / MNDWI / NBR / NDSI / EVI / SAVI bins, with published semantics. V19 has the inverse problem of V13 — token names (`axis_0`, `q_bin_03`) are abstract coordinates with no spectral anchor. The recipe-fit pair determines interpretability as much as the recipe alone.

d) V20 (MI-weighted bands): V20 inherits V1’s interpretability (per-band q-bin tokens are spectrally grounded) and adds a label-aware twist: high-MI bands emit more copies than low-MI bands, so the topic top-tokens concentrate on the discriminative subspectrum. F-13 SHAP under V20 therefore surfaces precisely the bands the label information is concentrated in — the interpretability is the same as V1’s, but the SHAP attributions are sharper because the LDA fit has fewer near-zero-contribution tokens to spread mass across. V20 should be considered the most interpretable of the new recipes: high label signal + grounded vocabulary + no codebook indirection.

e) F-22 counterfactual under the extension: We do not re-run F-22 for every new recipe in this draft; the V20 counterfactual L1 ranking under LDA is competitive with V12 (median L1 ≥ 15 on the labelled scenes where V20 has been profiled), suggesting that label-aware amplification produces topics that are also harder to perturb. A full F-22 ($\times 7$ new recipes $\times 6$ scenes) is left to a follow-up.

VI. RECOMMENDATION

For interpretability reporting, we recommend the following ordering:

- 1) **F-13 SHAP attribution.** Recipe-grounded, defensible against reviewer pressure, and produces explanations a domain expert can immediately query.
- 2) **F-22 counterfactual robustness.** Complements F-2/F-7 by ruling out trivial-overfit explanations.
- 3) **F-15 LLM-judge.** Only after vocabulary-size normalisation. Without normalisation it rewards trivial recipes.

The V13–V20 extension (§V) further suggests a recipe-as-interpretability-tier classification:

- **Tier I (physically-grounded):** V1, V7, V14, V18, V20. Top SHAP tokens map to a wavelength, absorption feature, scale band, manifold mode, or label-weighted band. Use for any publishable interpretability claim.
- **Tier II (semantically-grounded):** V8, V10, V15. Top tokens map to endmember labels, group labels, or vegetation indices.
- **Tier III (learnt-codebook-grounded):** V11, V12, V13, V17. Top tokens are codewords / GMM components / dictionary atoms. Interpretable only via a separate codebook-decoder visualisation.
- **Tier IV (abstract-coordinate):** V19. Topics point to UMAP coordinates with no spectral anchor. Use only as a sanity baseline.